

Лекция 9.

БЧХ-коды и РС-коды

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Риз, Солюмон → трансу Сигнатура $d_{\text{min}} = n - k + 1$

- 1 -
- 2 проверка на четность
- 3 Хэмминга
- 4 расшир. Хэмминга

$$H = \begin{bmatrix} 1 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 1 & 1 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} (7,4) \quad GF(p) \rightarrow GF(p^m)$$

$$GF(2) \quad 1+x+x^3 = p(x)$$

$$\rightarrow GF(2^3)$$

$$\begin{aligned} &1 \\ &x \\ &1+x \\ &x^2+1 \\ &x^2+x \\ &x^2+x+1 \end{aligned}$$

$$\begin{aligned} &\alpha^0 \quad 1 \\ &\alpha \quad x \\ &\alpha^2 \quad x^2 \\ &\alpha^3 \quad 1+x^2 \\ &\alpha^4 \quad x+x^2 \\ &\alpha^5 \quad 1+x+x^2 \\ &\alpha^6 \quad x+x^2 \end{aligned}$$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 \\ 0 & 1 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 1 & 1 & 1 \end{pmatrix}$$

$$(15, 7)$$

$$GF(2) \rightarrow GF(2^4)$$

$$H = (1 \alpha \alpha^2 \dots \alpha^{14})$$

$$p(x) = 1+x+x^4$$

$$c(x) + e(x) = b(x)$$

$$S = \sum_{i=0}^{n-1} b_i \cdot \alpha^i = \sum_{i=0}^{n-1} (c_i + e_i) \alpha^i = \sum_{i=0}^{n-1} c_i \alpha^i + \sum_{i=0}^{n-1} e_i \alpha^i = e(\alpha)$$

В положении i - оценка $b(x) = x^i, S = \alpha^i$

$$e(x) = x^i + x^j, i \neq j$$

$$s_i = \alpha^i + \alpha^j$$

$$\begin{cases} s_1 = \alpha^i + \alpha^j \\ s_2 = f(\alpha^i) + f(\alpha^j) \end{cases}$$

$$H = \begin{pmatrix} 1 & \alpha & \dots & \alpha^{14} \\ \beta_1 & \dots & \beta_{15} \end{pmatrix} \beta_i - GF(2^m)$$

$$f(x) = c \cdot x$$

$$GF(2)$$

$$f(x) = x^2$$

$$f(x^2) = f^2(x)$$

$$1^2 = 1 \quad 0^2 = 0$$

$$f(x) = x^3$$

$$H = \begin{pmatrix} 1 & \alpha & \alpha^2 & \alpha^3 & \alpha^4 & \alpha^5 & \alpha^6 & \alpha^7 & \alpha^8 & \alpha^9 & \alpha^{10} & \alpha^{11} & \alpha^{12} & \alpha^{13} & \alpha^{14} \\ 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} & 1 & \alpha^3 & \alpha^6 & \alpha^9 & \alpha^{12} \end{pmatrix}$$

i, j - позиции ошибок $x = \alpha^i \quad y = \alpha^j$

$$\begin{cases} s_1 = x + y \\ s_2 = x^3 + y^3 \end{cases}$$

при одной или двух ошибках $s_1 \neq 0$

$$\frac{x^3 + y^3}{x + y} = \frac{s_2}{s_1}$$

$$x^2 + y^2 + xy = \frac{s_2}{s_1}$$

$$s_1^2 + xy = \frac{s_2}{s_1}$$

$$\begin{cases} xy = \frac{s_2}{s_1} - s_1^2 \\ x + y = s_1 \end{cases}$$

$$\sigma_1 = s_1$$

$$\sigma_2 = \frac{s_2}{s_1} - s_1^2$$

$$GF(2) \rightarrow GF(2^4)$$

$$p(x) = 1 + x + x^4$$

сменено

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glou non

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x^0
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$$H = \begin{array}{cccccc} 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 1 & 1 \\ \hline 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 & 0 \\ \hline 1 & 0 & 0 & 0 & 1 & 1 \end{array} \rightarrow$$

$$\rightarrow \begin{array}{cccccccccccccccc} 1 & 1 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 1 \\ \hline \end{array}$$

$$S_1 = \alpha^2 + \alpha^9 = \alpha^4 \quad S_2 = \alpha^6 + \alpha^{12} = \alpha^4$$

$$\frac{\alpha^4}{\alpha^{11}} + (\alpha^4)^2 = \frac{1}{\alpha^7} + \alpha^7 = \alpha^8 + \alpha^7 = \frac{1 + \alpha^{14}}{\alpha^7} = \frac{\alpha^3}{\alpha^7} = \alpha^{-4}$$

$$\sigma_1 = \alpha^4 \quad \sigma_2 = \alpha^{11}$$

$$z^2 + \alpha^{11}z + \alpha^4 = \phi$$

$$\begin{aligned} \frac{1}{2} (\alpha^2)^2 + \alpha^{11} \cdot \alpha^2 + \alpha^4 &= \alpha^4 + \alpha^{13} + \alpha^{11} = \\ &= 1 + x + 1 + x^2 + x^3 + x + x^2 + x^3 = \phi \end{aligned}$$

$$\begin{aligned} (\alpha^9)^2 + \alpha^{11} \cdot \alpha^9 + \alpha^{11} &= \alpha^3 + \alpha^5 + \alpha^{11} = \\ &= x^3 + x + x^2 + x + x^2 + x^3 = \phi \end{aligned}$$